



TensorFlow Lite: An Open Source Deep Learning Framework For Handheld Devices

This article talks about the use of machine learning and deep learning models on handheld devices like mobile phones, Raspberry Pi, and microcontrollers

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Mobile devices are now an important part of our lives. Empowering them with machine learning (ML) models helps in creating more creative and real-time applications. TensorFlow Lite is a framework developed by Google that is open source, product-ready, and cross-platform. It is used to build and train ML and deep learning (DL) models on devices like mobile, embedded, and IoT devices.

TensorFlow is a specially designed framework that converts pre-trained models residing in TensorFlow to a specific format that is suitable for hand-held devices by considering speed and storage optimisation needs. This format is found to work well for mobile devices with Android, iOS, or Linux based embedded devices like microcontrollers or Raspberry Pi. Deployment of the pre-trained model on mobile devices makes it convenient to use them at any point of time.

Interfacing your model with your mobile device

Let us look at the features of ML and DL models that are important for interfacing with mobile devices.

1. Low latency. Deploying DL models on mobile devices makes it convenient to use irrespective of network connectivity, as the deployed model with suitable



interface eliminates a round-trip to the server and makes access of the functionality faster.

2. Light weight. Mobile devices have limited memory and computation power. Hence, the DL model deployed on a mobile device should be in smaller binary size and light-weight.

3. Safety. The interface made for the deployed model on the mobile device is accessible within the device only and it shares no data outside the device or network, eliminating data leak issues.

4. Limited power consumption. Connection to external network demands maximum power on the mobile device. However, models deployed on the hand-held device do not demand network connection, resulting in low power consumption.